

Haowen Zhang

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EDUCATION

University of California, San Diego, San Diego, CA
Bachelor of Science in Data Science / Joint Major Mathematics & Econ

September 2023 - June 2027
GPA: 3.877/4.00

RELEVANT TECHNICAL SKILLS

Computing Languages and Technologies - Proficient in using Python, Java, C++

Database Technologies - SQL, MySQL

WORK EXPERIENCE

Jiangsu Suli Environmental Technology Co., Ltd., Nanjing, Jiangsu — Data Analysis Engineer **July 2023 - August 2023**

- Built models to improve the accuracy of predicting the air quality in the future so that can take actions ahead of time
- Used linear regression, XGBoost, and SVM models to predict the content of PM2.5, PM10, SO2, and NO2 in the atmosphere based on past weather data.
- Used the R2 evaluation index to judge the accuracy of each of the models with real values.
- Used Pyecharts, Matplotlib to draw pollution line charts, pollution bar charts, and pollution maps.
- Used matplotlib to draw x Wind Rose and Pollution Rose Diagrams to help the company analyze the quality of the air. The graphs were used in the company's weekly presentation.

Publication

Published "Evaluating the quality of YouTube videos using sentiment analysis" paper on the Journal: "Advances in Computer and Communication" DOI: <http://dx.doi.org/10.26855/acc.2025.01.006> March 2025

- Used web scraping to fetch comments under YouTube videos and the number of views and likes.
- Used TextBlob and VADER sentiment analyzers to evaluate the comments
- Used mathematical statistics to analyze the scores given by TextBlob and VADER
- Used Excel and Matplotlib to draw graphs to analyze the trends

Projects

Built a recipes recommendation model to help customers find the recipes with high ratings March 2025

- Used "kinds of food used to make the dishes", "comments written by customers", and "nutrition content" to train Random Forest Model to predict the rating of recipes
- Used column transformers to increase the accuracy and precision of the model
- Used grid search to find the best hyperparameter of the model
- Built a fairness test to test whether our model is fair among different groups of data.

Built an n-gram model to predict the sentences based on words given February 2025

- Used web scraping to get examples of sentences from books
- Calculated the probability of a word existing before and after other words to estimate the sentences

An exploratory analysis of US Airbnb prices November 2025

- The columns are divided into groups (numerical, categorical, text) and cleaned.
- Correlation and types of missingness are analyzed to find out the relationship between the columns.
- Used Matplotlib to draw graphs to analyze the relationships between the columns in the datasets
- StandardScalar transformer is used for the numerical features, Onehotencoder is used for the categorical features, TF-IDF and SVD are used for the text features.
- Linear regression, ridge regression, XGBoost are used to predict the price of Airbnb.
- The results are analyzed by RMSE, MAE, R²

Leadership

Worked as Math Tutor and Grader September 2025 – December 2025

- Led discussion section to review the knowledge taught in the lectures, and guided the students do practice problems
- Holded office hours to discuss materials taught in the lectures
- Helped the TAs grade the exam papers
- Helped to proctor the exams